

Project Overview

This project aims to modernize a small library by creating a cohesive database system for the inventory and memberships. The database will store all items in the library, track transactions by clients, generate reports, generate fines, and maintain a record of other relevant information.

Scope

On the staff side, the system should be able to handle storage of a variety of media with associated information, track borrowing and return transactions, and also generate reports of trends, fines, and client activity. On the client side, the system will track ID, name, and contact information, and enforce membership fees/limits/etc.

Glossary

Platform

For our database we chose to use MySQL. We decided that MySQL would be the most convenient for the members to work on from home. We had some members who were familiar with MySQL and database design. Ease of use and the availability of documentation and tools also factored into our decision.

Physical Schema

```
-- =====
-- CREATE AND USE DATABASE
-- =====

DROP DATABASE IF EXISTS `bookstore_database`;
CREATE DATABASE `bookstore_database`
  DEFAULT CHARACTER SET utf8mb4
  COLLATE utf8mb4_0900_ai_ci;
USE `bookstore_database`;

-- =====
-- SESSION SETTINGS
-- =====

/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!50503 SET NAMES utf8mb4 */;
/*!40103 SET @OLD_TIME_ZONE=@@TIME_ZONE */;
/*!40103 SET TIME_ZONE='+00:00' */;
/*!40014 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0 */;
/*!40014 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS,
FOREIGN_KEY_CHECKS=0 */;
/*!40101 SET @OLD_SQL_MODE=@@SQL_MODE,
SQL_MODE='NO_AUTO_VALUE_ON_ZERO' */;
```

```
/*!40111 SET @OLD_SQL_NOTES=@@SQL_NOTES, SQL_NOTES=0 */;
```

```
-- =====  
-- TABLE create  
-- =====  
DROP TABLE IF EXISTS `staff`;  
CREATE TABLE `staff` (  
  `StaffID`      BIGINT UNSIGNED NOT NULL,  
  `Name`         varchar(50) NOT NULL,  
  `Age`          int          DEFAULT NULL,  
  `PhoneNumber`  varchar(20)   DEFAULT NULL,  
  `Address`      varchar(100) DEFAULT NULL,  
  `SSN`          varchar(20)  DEFAULT NULL,  
  `DOB`          date         DEFAULT NULL,  
  PRIMARY KEY (`StaffID`),  
  CONSTRAINT `staff_chk_1` CHECK (`Age` >= 0)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

```
DROP TABLE IF EXISTS `membership`;  
CREATE TABLE `membership` (  
  `MembershipTypeID` int NOT NULL,  
  `TypeName`         varchar(50) NOT NULL,  
  `BorrowLimit`      int NOT NULL DEFAULT 5,  
  `DailyFineRate`    decimal(5,2) NOT NULL DEFAULT 0.25,  
  PRIMARY KEY (`MembershipTypeID`),  
  CONSTRAINT `membership_chk_1` CHECK (`BorrowLimit` >= 0),  
  CONSTRAINT `membership_chk_2` CHECK (`DailyFineRate` >= 0)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

```
DROP TABLE IF EXISTS `client`;  
CREATE TABLE `client` (  
  `ClientID`      BIGINT UNSIGNED NOT NULL,  
  `Name`          varchar(50) NOT NULL,  
  `ContactInformation` varchar(100) DEFAULT NULL,  
  `Membership`     varchar(20) DEFAULT NULL,  
  `AccountStatus`  varchar(20) DEFAULT NULL,  
  `MembershipTypeID` int DEFAULT NULL,  
  PRIMARY KEY (`ClientID`),  
  KEY `fk_Client_Membership` (`MembershipTypeID`),  
  CONSTRAINT `fk_Client_Membership`  
    FOREIGN KEY (`MembershipTypeID`)  
    REFERENCES `membership` (`MembershipTypeID`)
```

```
        ON DELETE SET NULL
        ON UPDATE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

```
DROP TABLE IF EXISTS `copy`;
CREATE TABLE `copy` (
  `CopyID`          BIGINT UNSIGNED NOT NULL,
  `Type`            varchar(20) NOT NULL,
  `AvailabilityStatus` varchar(20) NOT NULL,
  PRIMARY KEY (`CopyID`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

```
DROP TABLE IF EXISTS `book`;
CREATE TABLE `book` (
  `ISBN`            varchar(20) NOT NULL,
  `Title`           varchar(100) NOT NULL,
  `Author`          varchar(100) NOT NULL,
  `Publisher`       varchar(100) DEFAULT NULL,
  `Genre`           varchar(50) DEFAULT NULL,
  `PublicationYear` int          DEFAULT NULL,
  `AvailabilityStatus` varchar(20) DEFAULT NULL,
  `CopyID`          BIGINT UNSIGNED DEFAULT NULL,
  PRIMARY KEY (`ISBN`),
  KEY `fk_Book_Copy` (`CopyID`),
  CONSTRAINT `fk_Book_Copy`
    FOREIGN KEY (`CopyID`)
      REFERENCES `copy` (`CopyID`)
      ON DELETE SET NULL
      ON UPDATE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

```
DROP TABLE IF EXISTS `digitalmedia`;
CREATE TABLE `digitalmedia` (
  `ID`             varchar(20) NOT NULL,
  `Title`          varchar(100) NOT NULL,
  `Creator`        varchar(50) DEFAULT NULL,
  `Format`         varchar(50) NOT NULL,
  `ReleaseYear`    int          DEFAULT NULL,
  `Genre`          varchar(50) DEFAULT NULL,
  `AvailabilityStatus` varchar(20) DEFAULT NULL,
  `CopyID`         BIGINT UNSIGNED DEFAULT NULL,
```

```

PRIMARY KEY (`ID`),
KEY `fk_Digital_Copy` (`CopyID`),
CONSTRAINT `fk_Digital_Copy`
    FOREIGN KEY (`CopyID`)
    REFERENCES `copy` (`CopyID`)
    ON DELETE SET NULL
    ON UPDATE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

```

```

DROP TABLE IF EXISTS `magazine`;
CREATE TABLE `magazine` (
    `ID`          varchar(50)  NOT NULL,
    `Title`       varchar(100) NOT NULL,
    `IssueNumber` varchar(20)  NOT NULL,
    `PublicationDate` date      DEFAULT NULL,
    `AvailabilityStatus` varchar(20) DEFAULT NULL,
    `CopyID`      BIGINT UNSIGNED DEFAULT NULL,
    PRIMARY KEY (`ID`),
    KEY `fk_Magazine_Copy` (`CopyID`),
    CONSTRAINT `fk_Magazine_Copy`
        FOREIGN KEY (`CopyID`)
        REFERENCES `copy` (`CopyID`)
        ON DELETE SET NULL
        ON UPDATE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

```

```

DROP TABLE IF EXISTS `borrowtransaction`;
CREATE TABLE `borrowtransaction` (
    `TransactionID` BIGINT UNSIGNED NOT NULL,
    `CheckoutDate`  date             NOT NULL,
    `DueDate`       date             NOT NULL,
    `ReturnDate`    date             DEFAULT NULL,
    `CopyID`        BIGINT UNSIGNED NOT NULL,
    `ClientID`      BIGINT UNSIGNED NOT NULL,
    `StaffID`       BIGINT UNSIGNED DEFAULT NULL,
    PRIMARY KEY (`TransactionID`),
    KEY `fk_Trans_Copy` (`CopyID`),
    KEY `fk_Trans_Client` (`ClientID`),
    KEY `fk_Trans_Staff` (`StaffID`),
    CONSTRAINT `fk_Trans_Client`
        FOREIGN KEY (`ClientID`)
        REFERENCES `client` (`ClientID`)

```

```

        ON DELETE RESTRICT
        ON UPDATE CASCADE,
CONSTRAINT `fk_Trans_Copy`
    FOREIGN KEY (`CopyID`)
    REFERENCES `copy` (`CopyID`)
    ON DELETE RESTRICT
    ON UPDATE CASCADE,
CONSTRAINT `fk_Trans_Staff`
    FOREIGN KEY (`StaffID`)
    REFERENCES `staff` (`StaffID`)
    ON DELETE SET NULL
    ON UPDATE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

```

```

DROP TABLE IF EXISTS `fine`;
CREATE TABLE `fine` (
    `FineID`      BIGINT UNSIGNED NOT NULL,
    `TransactionID` BIGINT UNSIGNED NOT NULL,
    `Amount`      decimal(5,2) NOT NULL,
    `PaymentDate` date          DEFAULT NULL,
    PRIMARY KEY (`FineID`),
    KEY `fk_Fine_Trans` (`TransactionID`),
    CONSTRAINT `fk_Fine_Trans`
        FOREIGN KEY (`TransactionID`)
        REFERENCES `borrowtransaction` (`TransactionID`)
        ON DELETE CASCADE
        ON UPDATE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

```

```

DROP TABLE IF EXISTS `report`;
CREATE TABLE `report` (
    `ReportID`      BIGINT UNSIGNED NOT NULL,
    `DateOfGeneration` date          NOT NULL,
    `Type`          varchar(20) DEFAULT NULL,
    `StaffID`       BIGINT UNSIGNED NOT NULL,
    PRIMARY KEY (`ReportID`),
    KEY `fk_Report_Staff` (`StaffID`),
    CONSTRAINT `fk_Report_Staff`
        FOREIGN KEY (`StaffID`)
        REFERENCES `staff` (`StaffID`)
        ON DELETE RESTRICT
        ON UPDATE CASCADE

```

```
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

```
DROP TABLE IF EXISTS `reservation`;
```

```
CREATE TABLE `reservation` (  
  `ReservationID` int NOT NULL,  
  `ClientID` BIGINT UNSIGNED NOT NULL,  
  `ItemID` BIGINT UNSIGNED NOT NULL,  
  `ReservationDate` date NOT NULL,  
  `Status` varchar(20) NOT NULL,  
  PRIMARY KEY (`ReservationID`),  
  KEY `fk_Res_Client` (`ClientID`),  
  KEY `fk_Res_Copy` (`ItemID`),  
  CONSTRAINT `fk_Res_Client`  
    FOREIGN KEY (`ClientID`)  
    REFERENCES `client` (`ClientID`)  
    ON DELETE CASCADE  
    ON UPDATE CASCADE,  
  CONSTRAINT `fk_Res_Copy`  
    FOREIGN KEY (`ItemID`)  
    REFERENCES `copy` (`CopyID`)  
    ON DELETE RESTRICT  
    ON UPDATE CASCADE
```

```
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

Data Population

USE bookstore_database;

```
INSERT IGNORE INTO membership (MembershipTypeID, TypeName, BorrowLimit,
DailyFineRate) VALUES
```

```
(0, 'A', 5, 0.00),
(1, 'B', 3, 0.25),
(2, 'C', 1, 1.00);
```

```
INSERT IGNORE INTO staff (StaffID, Name, Age, PhoneNumber, Address, SSN, DOB)
VALUES
```

```
(277043089914, 'Michelle Wallace', 55, '(001) 454-3910', '64482 Amanda Loop, Figueroaview,
NV 49234', '582-81-2258', STR_TO_DATE('12/24/1969', '%m/%d/%Y')),
(384994027174, 'Bryan Morales', 65, '(284) 576-3778', '709 Anthony Mountains, Onealtown, NC
09211', '704-11-6474', STR_TO_DATE('09/24/1959', '%m/%d/%Y')),
(192707852656, 'Brian Smith', 63, '(432) 943-1675', '72852 Renee Mills Suite 667,
Blairborough, IL 77378', '728-07-6363', STR_TO_DATE('07/23/1961', '%m/%d/%Y')),
(482225930755, 'Bradley Lopez', 39, '(239) 491-8456', '774 Lewis Grove Apt. 002, Adamtown,
CA 56566', '697-29-9972', STR_TO_DATE('04/11/1986', '%m/%d/%Y')),
(847157462649, 'Connie Lee', 62, '(119) 155-8754', '533 Chambers Street, Larryfurt, NJ 45581',
'893-92-3932', STR_TO_DATE('04/28/1962', '%m/%d/%Y')),
(655919975680, 'Juan Smith', 52, '(706) 777-1113', '8602 Tracy Crossroad, Sarahmouth, CT
73515', '293-79-8865', STR_TO_DATE('12/25/1972', '%m/%d/%Y')),
(18984280546, 'Paula Palmer', 21, '(844) 301-9126', '886 David Rue, Davidshire, IN 88959',
'734-91-8957', STR_TO_DATE('03/31/2004', '%m/%d/%Y')),
(841606709552, 'Patrick Valenzuela', 56, '(845) 357-4922', '9026 Theresa Pass Apt. 237, East
Jessica, IN 50912', '362-78-9684', STR_TO_DATE('04/24/1968', '%m/%d/%Y')),
(63911658753, 'Stephanie Rodriguez', 35, '(895) 774-0382', '785 Jonathon Rapids Suite 354,
Acevedoborough, MD 95719', '766-32-5623', STR_TO_DATE('04/25/1989', '%m/%d/%Y')),
(347903932193, 'Mary Perez', 39, '(240) 598-0250', '51056 Patricia Forge, Grahamstad, HI
85576', '337-94-6696', STR_TO_DATE('05/02/1985', '%m/%d/%Y'));
```

```
INSERT IGNORE INTO client (ClientID, Name, ContactInformation, Membership,
AccountStatus, MembershipTypeID) VALUES
```

```
(779102014641, 'Norma Fisher', '(876) 475-9382', 'C', 'Inactive', 2),
(162788563211, 'Danielle Browning', '(001) 924-1157', 'C', 'Warning', 2),
(392888992260, 'Christine Tran', '(840) 801-6097', 'C', 'Suspended', 2),
(926509076716, 'Amy Stark', '(328) 711-5871', 'A', 'Warning', 0),
(777315144757, 'Eddie Martinez', NULL, 'A', 'Active', 0),
(670019806533, 'Thomas Harris', '(189) 471-9659', 'C', 'Warning', 2),
(106325369465, 'Amanda Zavala', '(711) 220-1868', 'A', 'Inactive', 0),
(862536141388, 'Stephanie Lawrence', '(947) 751-5917', 'A', 'Active', 0),
```

(908358000255, 'Autumn Robinson', '(560) 123-0989', 'C', 'Suspended', 2),
(664904820126, 'Alyssa Reid', '(191) 615-1090', 'C', 'Warning', 2),
(242955608655, 'Robert Giles', '(869) 141-3145', 'A', 'Inactive', 0),
(98844556456, 'Antonio Mcdonald', '(116) 345-7923', 'C', 'Suspended', 2),
(600450083709, 'Autumn Morse', '(120) 769-8456', 'B', 'Suspended', 1),
(72458122216, 'Angel Campbell', '(759) 459-9246', 'C', 'Warning', 2),
(368798545706, 'Sandra Miller', NULL, 'C', 'Active', 2),
(345462137506, 'Melissa Moore', '(769) 606-9602', 'C', 'Warning', 2),
(534771852898, 'David Moore', '(789) 007-5470', 'A', 'Active', 0),
(588022882666, 'Mrs. Meghan Stewart', '(008) 913-1934', 'A', 'Inactive', 0),
(386979565044, 'William Garcia', '(104) 714-2851', 'A', 'Warning', 0),
(609741931457, 'Michael Grant', '(909) 776-5823', 'A', 'Active', 0),
(963447298338, 'Chris Cook', '(159) 004-2294', 'C', 'Inactive', 2),
(152491468918, 'Brian Cross', NULL, 'C', 'Active', 2),
(65543315395, 'Elizabeth Vincent', NULL, 'C', 'Inactive', 2),
(151534339699, 'Elizabeth Houston', '(611) 877-5517', 'C', 'Active', 2),
(363714941482, 'Nicholas Kline', '(961) 113-3060', 'B', 'Active', 1),
(100708262468, 'Justin Roberts', NULL, 'A', 'Inactive', 0),
(799911506282, 'Rachel Klein', '(615) 349-2635', 'A', 'Active', 0),
(331180881681, 'Rodney Farrell', '(873) 176-4303', 'C', 'Active', 2),
(616208160855, 'Ashley Ellis', '(821) 972-9668', 'A', 'Suspended', 0),
(487380045830, 'Miss Shelly Daniels DVM', '(555) 082-4926', 'B', 'Active', 1),
(424533559245, 'Michelle Roman', '(180) 132-0407', 'C', 'Warning', 2),
(690064463489, 'Christopher Hernandez', '(175) 868-8091', 'A', 'Active', 0),
(774490855370, 'Steven Butler', '(699) 300-2489', 'A', 'Active', 0),
(878829203004, 'Terri Turner', '(660) 223-4500', 'B', 'Inactive', 1),
(320195254711, 'Megan Hernandez', '(609) 767-0172', 'C', 'Suspended', 2),
(240123516435, 'Lorraine Garcia', '(153) 671-0979', 'C', 'Inactive', 2),
(935868384284, 'Tracy Reeves', '(264) 183-0675', 'A', 'Active', 0),
(158774376402, 'Stephanie Williams', '(089) 933-1886', 'A', 'Inactive', 0),
(136178644349, 'Steven Henderson', '(161) 162-0766', 'B', 'Active', 1),
(978212965548, 'Thomas Rivers', '(001) 415-1150', 'B', 'Inactive', 1),
(227259652028, 'Briana Smith', NULL, 'C', 'Suspended', 2),
(80444014843, 'Monique Santana', '(001) 045-0932', 'C', 'Warning', 2),
(881493143287, 'Daniel Rogers', '(485) 291-1894', 'B', 'Warning', 1),
(702703408825, 'Emily Sanchez', NULL, 'C', 'Active', 2),
(981758150271, 'Donna Barnes', '(290) 611-2648', 'A', 'Warning', 0),
(44756014165, 'Timothy Garza', '(513) 758-0660', 'C', 'Warning', 2),
(540877402763, 'Tristan Davis', NULL, 'B', 'Warning', 1),
(913899456057, 'Brian Rodriguez', '(127) 626-6237', 'C', 'Warning', 2),
(574947128222, 'Mark Webb', '(407) 970-3414', 'C', 'Inactive', 2),
(108706256089, 'Brooke Mccarthy', '(173) 864-0140', 'B', 'Suspended', 1);

INSERT IGNORE INTO copy (CopyID, Type, AvailabilityStatus) VALUES

(491485519980, 'Book', 'Available'),
(323189472559, 'Book', 'Available'),
(771900436262, 'Book', 'Available'),
(658393445831, 'Book', 'Available'),
(778084897890, 'Book', 'Available'),
(139441916268, 'Book', 'Available'),
(411437495765, 'Book', 'Available'),
(984901710141, 'Book', 'Available'),
(946002405289, 'Book', 'Available'),
(106071650495, 'Book', 'Available'),
(354023658054, 'Book', 'Available'),
(565393293913, 'Book', 'Available'),
(426357542125, 'Book', 'Available'),
(488808576803, 'Book', 'Available'),
(767151867182, 'Book', 'Available'),
(925634441473, 'Book', 'Available'),
(785782374802, 'Book', 'Available'),
(530239166629, 'Book', 'Available'),
(705529635231, 'Book', 'Available'),
(550685652433, 'Book', 'Available'),
(61283062945223747482, 'Magazine', 'Available'),
(18020094731993304080, 'Magazine', 'Available'),
(83748304109018074095, 'Magazine', 'Available'),
(25129439863619128378, 'Magazine', 'Available'),
(10752600121128046276, 'Magazine', 'Available'),
(34664092976165992674, 'Magazine', 'Available'),
(64132539988041044087, 'Magazine', 'Available'),
(95130706565765160262, 'Magazine', 'Available'),
(29972608590661967705, 'Magazine', 'Available'),
(30460281374294866211, 'Magazine', 'Available'),
(14037923478426030347, 'Magazine', 'Available'),
(41614077772381461334, 'Magazine', 'Available'),
(84094843450387855609, 'Magazine', 'Available'),
(49268086983727935119, 'Magazine', 'Available'),
(43507434037019985963, 'Magazine', 'Available'),
(99134704012614036042, 'Magazine', 'Available'),
(26004191332960245805, 'Magazine', 'Available'),
(50579887656586861483, 'Magazine', 'Available'),
(47648480152875188180, 'Magazine', 'Available'),
(46493745440291072166, 'Magazine', 'Available'),
(730795000000, 'DigitalMedia', 'Available'),
(232557000000, 'DigitalMedia', 'Available'),

(31515362208, 'DigitalMedia', 'Available'),
(963484000000, 'DigitalMedia', 'Available'),
(213939000000, 'DigitalMedia', 'Available'),
(770388000000, 'DigitalMedia', 'Available'),
(91099697472, 'DigitalMedia', 'Available'),
(301162000000, 'DigitalMedia', 'Available'),
(626674000000, 'DigitalMedia', 'Available'),
(497119000000, 'DigitalMedia', 'Available'),
(790022000000, 'DigitalMedia', 'Available'),
(171753000000, 'DigitalMedia', 'Available'),
(636887000000, 'DigitalMedia', 'Available'),
(935982000000, 'DigitalMedia', 'Available'),
(433216000000, 'DigitalMedia', 'Available'),
(101184000000, 'DigitalMedia', 'Available'),
(337415000000, 'DigitalMedia', 'Available'),
(653070000000, 'DigitalMedia', 'Available'),
(50304107602, 'DigitalMedia', 'Available'),
(444570000000, 'DigitalMedia', 'Available'),
(171365000000, 'DigitalMedia', 'Available'),
(405912000000, 'DigitalMedia', 'Available'),
(262956000000, 'DigitalMedia', 'Available');

INSERT IGNORE INTO book (ISBN, Title, Author, Publisher, Genre, PublicationYear, AvailabilityStatus, CopyID) VALUES
(1273754815743, 'Shadows of Eternity', 'Elena Winters', 'Silver Quill Press', 'Fantasy', 1974, 'No', 491485519980),
(3262456709743, 'The Crystal Code', 'Marcus Thorn', 'Midnight Lantern Publishing', 'Fiction', 1980, 'No', 323189472559),
(4887721181798, 'Whispers in the Wind', 'Isabella Drake', 'Starlight House', 'Biography', 1968, 'No', 771900436262),
(4029656002307, 'Echoes of Tomorrow', 'Julian Frost', 'Ironbound Books', 'Non-Fiction', 1967, 'Yes', 658393445831),
(5421207680208, 'Beneath the Iron Sky', 'Natalie Crowe', 'Echo Peak Publishing', 'Fiction', 2012, 'Yes', 778084897890),
(8049194392752, 'The Forgotten Kingdom', 'Damien Hale', 'Celestial Ink', 'Mystery', 2010, 'No', 139441916268),
(7721818335132, 'Realm of Secrets', 'Lydia Blackwood', 'Whispering Pines Press', 'Fantasy', 2024, 'No', 411437495765),
(5665528842083, 'Chronicles of the Void', 'Connor Vance', 'Emberlight Publications', 'History', 1997, 'Yes', 984901710141),
(6030091410282, 'The Last Ember', 'Seraphina Moore', 'Northern Gate Books', 'Biography', 1966, 'Yes', 946002405289),

(9158736461602, 'Dreams of Glass', 'Gideon Stroud', 'Obsidian Scroll', 'Non-Fiction', 1984, 'No', 106071650495),
(6796154824223, 'Path of the Ancients', 'Aria Tempest', 'Radiant Leaf Press', 'Romance', 1987, 'No', 354023658054),
(672027792524, 'Voices from the Deep', 'Victor Stone', 'Twilight Beacon', 'Romance', 1974, 'No', 565393293913),
(3233694870907, 'Labyrinth of Souls', 'Mira Langley', 'Stormwatch Publications', 'Biography', 1972, 'No', 426357542125),
(1023832279531, 'Twilight Covenant', 'Orion Black', 'Wandering Word Works', 'Mystery', 2011, 'No', 488808576803),
(6851259388489, 'The Ninth Horizon', 'Clara Voss', 'Sapphire Tome House', 'History', 2024, 'No', 767151867182),
(8769680377543, 'Oath of Ashes', 'Rowan Vale', 'Phoenix Crest Press', 'Fantasy', 1986, 'No', 925634441473),
(5407350920242, 'Sands of Time', 'Ivy Nightshade', 'Golden Atlas Publishing', 'History', 1981, 'No', 785782374802),
(5976911093807, 'Frozen Reverie', 'Dorian Cross', 'Moonrise Editions', 'Science Fiction', 2022, 'No', 530239166629),
(3621070726942, 'The Hollow Crown', 'Evelyn Storm', 'Crimson Hearth Books', 'Romance', 1996, 'No', 705529635231),
(6684668056401, 'Legacy of the Stars', 'Tristan Vale', 'Zephyrbound Press', 'Fantasy', 1963, 'Yes', 550685652433);

INSERT IGNORE INTO magazine (ID, Title, IssueNumber, PublicationDate, AvailabilityStatus, CopyID) VALUES

('91364862083766535851', 'Car and Driver', '831', '2015-03-16', 'Available', 61283062945223747482),
('24452447094200899317', 'Motor Trend', '918', '2006-08-18', 'Available', 18020094731993304080),
('90639621556383653871', 'Top Gear Magazine', '336', '2013-04-01', 'Available', 83748304109018074095),
('11050203816882597162', 'Autocar', '822', '1998-03-16', 'Available', 25129439863619128378),
('16596213807127021220', 'Road & Track', '796', '2007-09-16', 'Available', 10752600121128046276),
('14131106827042409105', 'TIME', '72', '2024-02-01', 'Available', 34664092976165992674),
('75212912953541359066', 'The Economist', '142', '2024-03-01', 'Available', 64132539988041044087),
('90877976502182879450', 'Newsweek', '948', '2024-04-01', 'Available', 95130706565765160262),
('16802116532868856372', 'Forbes', '579', '2024-05-01', 'Available', 29972608590661967705),
('31433342416844503857', 'The New Yorker', '280', '2024-06-01', 'Available', 30460281374294866211),

('62101756267638911122', 'Scientific American', '400', '2024-03-05', 'Available',
14037923478426030347),
('14826729814846290022', 'Nature', '673', '2024-02-02', 'Available', 41614077772381461334),
('32665537064772465207', 'New Scientist', '751', '2020-01-11', 'Available',
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('12663784891363892912', 'National Geographic', '940', '2014-05-14', 'Available',
49268086983727935119),
('55546063833112115344', 'Popular Science', '640', '2012-11-01', 'Available',
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('87902198241574797489', 'Vogue', '881', '2021-02-05', 'Available', 99134704012614036042),
('53382725813343369496', 'Elle', '316', '2000-12-01', 'Available', 26004191332960245805),
('77576093034374493339', 'Harper's Bazaar', '301', '2023-07-12', 'Available',
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('66511524920328405331', 'GQ', '993', '2024-11-01', 'Available', 47648480152875188180),
('77560604328349747458', 'W Magazine', '961', '2024-02-17', 'Available',
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INSERT IGNORE INTO digitalmedia (ID, Title, Creator, Format, ReleaseYear, Genre,
AvailabilityStatus, CopyID) VALUES

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('230129000000', 'Saved by the Bell', 'Cairistiona Abbie', 'DVD', 1994, 'Children', 'no',
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('85913602825', 'Predictions of Fire', 'Paulina Haward', 'DVD', 1992, 'Documentary', 'no',
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('707642000000', 'King of Boys', 'Kemi Adetiba', 'DVD', 2018, 'Drama', 'yes', 213939000000),
('704559000000', 'The World's Most Amazing Vacation Rentals', 'Engelbert Shenley', 'DVD',
2021, 'Reality', 'no', 770388000000),
('435700000000', 'Tobot Galaxy Detectives', 'Nikoletta Connealy', 'DVD', 2019, 'Children', 'no',
91099697472),
('227710000000', 'La casa de papel', 'Paule Rodell', 'Blu-ray', 2021, 'Crime', 'yes',
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('837187000000', 'A Cinderella Story', 'Mark Rosman', 'Blu-ray', 2004, 'Children', 'no',
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('669280000000', 'Blade Runner: The Final Cut', 'Ridley Scott', 'DVD', 1982, 'Action', 'yes',
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('558347000000', 'Green Lantern', 'Martin Campbell', 'Blu-ray', 2011, 'Sci-Fi', 'yes',
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('618148000000', 'Gridlock'd', 'Merwyn Harriagn', 'DVD', 1986, 'Crime', 'yes', 171753000000),
('83218955767', 'Level 16', 'Danishka Esterhazy', 'DVD', 2018, 'Sci-Fi', 'yes', 636887000000),
('69015153056', 'Marshall', 'Reginald Hudlin', 'Blu-ray', 2017, 'Drama', 'yes', 935982000000),
('678339000000', 'The Howling', 'Somerset Goly', 'DVD', 1992, 'Horror', 'no', 433216000000),

('982096000000', 'Bob Ross: Happy Accidents', 'Betrayal & Greed', 'Blu-ray', 2021,
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 ('395777000000', 'Hit & Run', 'Moe Hoyer', 'Blu-ray', 2021, 'Mystery', 'no', 337415000000),
 ('983241000000', 'The Englishman', 'Carina Ciobutaru', 'Blu-ray', 2007, 'Romance', 'no',
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 ('324128000000', 'Rancho Notorious', 'Mella Janaushek', 'Blu-ray', 2009, 'Western', 'no',
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 ('675201000000', 'Ferris Bueller's Day Off', 'John Hughes', 'Blu-ray', 1986, 'Comedy', 'no',
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 ('935952000000', 'Inception', 'Christopher Nolan', 'DVD', 2010, 'Thrillers', 'yes', 405912000000),
 ('496599000000', 'School of Rock', 'Richard Linklater', 'DVD', 2003, 'Comedy', 'yes',
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INSERT IGNORE INTO borrowtransaction (TransactionID, CheckoutDate, DueDate,
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 (1004, STR_TO_DATE('01/22/2024', '%m/%d/%Y'), STR_TO_DATE('02/05/2024',
 '%m/%d/%Y'), STR_TO_DATE('02/01/2024', '%m/%d/%Y'), 18020094731993304080,
 926509076716, 192707852656),
 (1005, STR_TO_DATE('01/25/2024', '%m/%d/%Y'), STR_TO_DATE('02/08/2024',
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 (1006, STR_TO_DATE('01/27/2024', '%m/%d/%Y'), STR_TO_DATE('02/10/2024',
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 (1007, STR_TO_DATE('01/30/2024', '%m/%d/%Y'), STR_TO_DATE('02/13/2024',
 '%m/%d/%Y'), STR_TO_DATE('02/12/2024', '%m/%d/%Y'), 31515362208, 106325369465,
 655919975680),
 (1008, STR_TO_DATE('02/02/2024', '%m/%d/%Y'), STR_TO_DATE('02/16/2024',
 '%m/%d/%Y'), NULL, 963484000000, 862536141388, 18984280546),
 (1009, STR_TO_DATE('02/05/2024', '%m/%d/%Y'), STR_TO_DATE('02/19/2024',
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 (1010, STR_TO_DATE('02/07/2024', '%m/%d/%Y'), STR_TO_DATE('02/21/2024',
 '%m/%d/%Y'), STR_TO_DATE('02/20/2024', '%m/%d/%Y'), 770388000000, 664904820126,
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 (1011, STR_TO_DATE('03/01/2025', '%m/%d/%Y'), STR_TO_DATE('03/15/2025', '%m/%d/%Y'),
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(1013, STR_TO_DATE('03/05/2025','%m/%d/%Y'), STR_TO_DATE('03/19/2025','%m/%d/%Y'),
NULL , 771900436262 , 600450083709 , 277043089914),
(1014, STR_TO_DATE('03/07/2025','%m/%d/%Y'), STR_TO_DATE('03/21/2025','%m/%d/%Y'),
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(1015, STR_TO_DATE('03/09/2025','%m/%d/%Y'), STR_TO_DATE('03/23/2025','%m/%d/%Y'),
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(1016, STR_TO_DATE('03/11/2025','%m/%d/%Y'), STR_TO_DATE('03/25/2025','%m/%d/%Y'),
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(1018, STR_TO_DATE('03/15/2025','%m/%d/%Y'), STR_TO_DATE('03/29/2025','%m/%d/%Y'),
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(1019, STR_TO_DATE('03/17/2025','%m/%d/%Y'), STR_TO_DATE('03/31/2025','%m/%d/%Y'),
STR_TO_DATE('03/28/2025','%m/%d/%Y'), 106071650495 , 386979565044 , 277043089914),
(1020, STR_TO_DATE('03/19/2025','%m/%d/%Y'), STR_TO_DATE('04/02/2025','%m/%d/%Y'),
STR_TO_DATE('03/30/2025','%m/%d/%Y'), 354023658054 , 609741931457 , 384994027174),
(1021, STR_TO_DATE('03/21/2025','%m/%d/%Y'), STR_TO_DATE('04/04/2025','%m/%d/%Y'),
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(1022, STR_TO_DATE('03/23/2025','%m/%d/%Y'), STR_TO_DATE('04/06/2025','%m/%d/%Y'),
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STR_TO_DATE('04/05/2025','%m/%d/%Y'), 488808576803 , 65543315395 , 277043089914),
(1024, STR_TO_DATE('03/27/2025','%m/%d/%Y'), STR_TO_DATE('04/10/2025','%m/%d/%Y'),
STR_TO_DATE('04/07/2025','%m/%d/%Y'), 767151867182 , 151534339699 , 384994027174),
(1025, STR_TO_DATE('03/29/2025','%m/%d/%Y'), STR_TO_DATE('04/12/2025','%m/%d/%Y'),
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INSERT INTO fine (FineID, TransactionID, Amount, PaymentDate) VALUES

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(101, 1003, 5.75, STR_TO_DATE('05/20/2023', '%m/%d/%Y')),
(102, 1005, 3.25, STR_TO_DATE('05/13/2023', '%m/%d/%Y')),
(103, 1009, 7.50, STR_TO_DATE('11/18/2023', '%m/%d/%Y')),
(104, 1001, 0.00, STR_TO_DATE('03/01/2024', '%m/%d/%Y')),
(105, 1002, 0.00, STR_TO_DATE('05/27/2024', '%m/%d/%Y')),
(106, 1004, 0.00, STR_TO_DATE('10/25/2024', '%m/%d/%Y')),
(107, 1006, 0.00, STR_TO_DATE('01/18/2025', '%m/%d/%Y')),
(108, 1007, 0.00, STR_TO_DATE('03/30/2025', '%m/%d/%Y')),
(109, 1008, 0.00, STR_TO_DATE('04/16/2025', '%m/%d/%Y')),
(110, 1010, 0.00, STR_TO_DATE('02/28/2025', '%m/%d/%Y'));

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INSERT IGNORE INTO reservation (ReservationID, ClientID, ItemID, ReservationDate, Status)
VALUES

(501, 779102014641, 778084897890, STR_TO_DATE('01/10/2024', '%m/%d/%Y'), 'Fulfilled'),
(502, 162788563211, 925634441473, STR_TO_DATE('01/12/2024', '%m/%d/%Y'), 'Fulfilled'),
(503, 392888992260, 61283062945223747482, STR_TO_DATE('01/15/2024', '%m/%d/%Y'),
'Fulfilled'),
(504, 926509076716, 18020094731993304080, STR_TO_DATE('01/17/2024', '%m/%d/%Y'),
'Fulfilled'),
(505, 777315144757, 730795000000, STR_TO_DATE('01/20/2024', '%m/%d/%Y'), 'Fulfilled'),
(506, 670019806533, 232557000000, STR_TO_DATE('01/22/2024', '%m/%d/%Y'), 'Fulfilled'),
(507, 106325369465, 31515362208, STR_TO_DATE('01/25/2024', '%m/%d/%Y'), 'Fulfilled'),
(508, 862536141388, 963484000000, STR_TO_DATE('01/28/2024', '%m/%d/%Y'), 'Fulfilled'),
(509, 908358000255, 213939000000, STR_TO_DATE('01/30/2024', '%m/%d/%Y'), 'Fulfilled'),
(510, 664904820126, 770388000000, STR_TO_DATE('02/02/2024', '%m/%d/%Y'), 'Fulfilled');

INSERT IGNORE INTO report (ReportID, DateOfGeneration, Type, StaffID) VALUES
(81326531512, '2016-05-12', 'revenue', 277043089914),
(990121808357, '2021-03-06', 'revenue', 384994027174),
(290084112145, '2019-09-24', 'revenue', 482225930755),
(941415280066, '2001-03-17', 'monthly', 192707852656),
(521230953672, '2006-07-24', 'popularity', 847157462649);

Table Contents

1 • SELECT * FROM staff

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	StaffID	Name	Age	PhoneNumber	Address	SSN	DOB
▶	18984280546	Paula Palmer	21	(844) 301-9126	886 David Rue, Davidshire, IN 88959	734-91-8957	2004-03-31
	63911658753	Stephanie Rodriguez	35	(895) 774-0382	785 Jonathon Rapids Suite 354, Acevedoborou...	766-32-5623	1989-04-25
	192707852656	Brian Smith	63	(432) 943-1675	72852 Renee Mills Suite 667, Blairborough, IL 7...	728-07-6363	1961-07-23
	277043089914	Michelle Wallace	55	(001) 454-3910	64482 Amanda Loop, Figueroaview, NV 49234	582-81-2258	1969-12-24
	347903932193	Mary Perez	39	(240) 598-0250	51056 Patricia Forge, Grahamstad, HI 85576	337-94-6696	1985-05-02
	384994027174	Bryan Morales	65	(284) 576-3778	709 Anthony Mountains, Onealtown, NC 09211	704-11-6474	1959-09-24
	482225930755	Bradley Lopez	39	(239) 491-8456	774 Lewis Grove Apt. 002, Adamtown, CA 56566	697-29-9972	1986-04-11
	655919975680	Juan Smith	52	(706) 777-1113	8602 Tracy Crossroad, Sarahmouth, CT 73515	293-79-8865	1972-12-25
	841606709552	Patrick Valenzuela	56	(845) 357-4922	9026 Theresa Pass Apt. 237, East Jessica, IN 5...	362-78-9684	1968-04-24
	847157462649	Connie Lee	62	(119) 155-8754	533 Chambers Street, Larryfurt, NJ 45581	893-92-3932	1962-04-28

1 • SELECT * FROM reservation

2

Result Grid

Filter Rows:

Edit:

Export/Import:

	ReservationID	ClientID	ItemID	ReservationDate	Status
▶	501	779102014641	778084897890	2024-01-10	Fulfilled
	502	162788563211	925634441473	2024-01-12	Fulfilled
	503	392888992260	18446744073709551615	2024-01-15	Fulfilled
	504	926509076716	18020094731993304080	2024-01-17	Fulfilled
	505	777315144757	730795000000	2024-01-20	Fulfilled
	506	670019806533	232557000000	2024-01-22	Fulfilled
	507	106325369465	31515362208	2024-01-25	Fulfilled
	508	862536141388	963484000000	2024-01-28	Fulfilled
	509	908358000255	213939000000	2024-01-30	Fulfilled
	510	664904820126	770388000000	2024-02-02	Fulfilled

1 • `SELECT * FROM report`

2

Result Grid			Filter Rows:	Edit:   
ReportID	DateOfGeneration	Type	StaffID	
81326531512	2016-05-12	revenue	277043089914	
290084112145	2019-09-24	revenue	482225930755	
521230953672	2006-07-24	popularity	847157462649	
941415280066	2001-03-17	monthly	192707852656	
990121808357	2021-03-06	revenue	384994027174	

1 • `SELECT * FROM membership`

2

Result Grid			Filter Rows:	Edit:   
MembershipTypeID	TypeName	BorrowLimit	DailyFineRate	
0	A	5	0.00	
1	B	3	0.25	
2	C	1	1.00	

1 • SELECT * FROM magazine

2

<						
Result Grid						
Filter Rows:						
Edit:						
Export/Import:						
Wrap Cell Content:						
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14131106827042409105	TIME	72	2024-02-01	Available	18446744073709551615	
14826729814846290022	Nature	673	2024-02-02	Available	18446744073709551615	
16596213807127021220	Road & Track	796	2007-09-16	Available	10752600121128046276	
16802116532868856372	Forbes	579	2024-05-01	Available	18446744073709551615	
24452447094200899317	Motor Trend	918	2006-08-18	Available	18020094731993304080	
31433342416844503857	The New Yorker	280	2024-06-01	Available	18446744073709551615	
32665537064772465207	New Scientist	751	2020-01-11	Available	18446744073709551615	
53382725813343369496	Elle	316	2000-12-01	Available	18446744073709551615	
55546063833112115344	Popular Science	640	2012-11-01	Available	18446744073709551615	
62101756267638911122	Scientific American	400	2024-03-05	Available	14037923478426030347	
66511524920328405331	GQ	993	2024-11-01	Available	18446744073709551615	
75212912953541359066	The Economist	142	2024-03-01	Available	18446744073709551615	
77560604328349747458	W Magazine	961	2024-02-17	Available	18446744073709551615	
77576093034374493339	Harper's Bazaar	301	2023-07-12	Available	18446744073709551615	
87902198241574797489	Vogue	881	2021-02-05	Available	18446744073709551615	
90639621556383653871	Top Gear Magazine	336	2013-04-01	Available	18446744073709551615	
90877976502182879450	Newsweek	948	2024-04-01	Available	18446744073709551615	
91364862083766535851	Car and Driver	831	2015-03-16	Available	18446744073709551615	

1 • SELECT * FROM digitalmedia

2

Result Grid							
Filter Rows: Edit: Export/Import: Wrap Cell Content:							
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227710000000	La casa de papel	Paule Rodell	Blu-ray	2021	Crime	yes	301162000000
230129000000	Saved by the Bell	Cairistiona Abbie	DVD	1994	Children	no	31515362208
324128000000	Rancho Notorious	Mella Janaushek	Blu-ray	2009	Western	no	50304107602
395777000000	Hit & Run	Moe Hoyes	Blu-ray	2021	Mystery	no	337415000000
435700000000	Tobot Galaxy Detectives	Nikoletta Connealy	DVD	2019	Children	no	91099697472
496599000000	School of Rock	Richard Linklater	DVD	2003	Comedy	yes	262956000000
539922000000	Beowulf	Robert Zemeckis	DVD	2007	Fantasy	no	444570000000
558347000000	Green Lantern	Martin Campbell	Blu-ray	2011	Sci-Fi	yes	790022000000
618148000000	Gridlock'd	Merwyn Harriagn	DVD	1986	Crime	yes	171753000000
669280000000	Blade Runner: The Final Cut	Ridley Scott	DVD	1982	Action	yes	497119000000
675201000000	Ferris Bueller's Day Off	John Hughes	Blu-ray	1986	Comedy	no	171365000000
678339000000	The Howling	Somerset Goly	DVD	1992	Horror	no	433216000000
69015153056	Marshall	Reginald Hudlin	Blu-ray	2017	Drama	yes	935982000000
704559000000	The World's Most Amazing ...	Engelbert Shenley	DVD	2021	Reality	no	770388000000
707642000000	King of Boys	Kemi Adetiba	DVD	2018	Drama	yes	213939000000
726774000000	Paranoia	Robert Luketic	DVD	2013	Thrillers	no	730795000000
83218955767	Level 16	Danishka Esterhazy	DVD	2018	Sci-Fi	yes	636887000000
837187000000	A Cinderella Story	Mark Rosman	Blu-ray	2004	Children	no	626674000000
85913602825	Predictions of Fire	Paulina Haward	DVD	1992	Docume...	no	963484000000
894366000000	Jaws	Steven Spielberg	Blu-ray	1975	Action	no	232557000000
935952000000	Inception	Christopher Nolan	DVD	2010	Thrillers	yes	405912000000
982096000000	Bob Ross: Happy Accidents	Betrayal & Greed	Blu-ray	2021	Docume...	yes	101184000000
983241000000	The Englishman	Carina Ciobutaru	Blu-ray	2007	Romance	no	653070000000

1 • `SELECT * FROM copy`

2

Result Grid			
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50304107602	DigitalMedia	Available	
91099697472	DigitalMedia	Available	
101184000000	DigitalMedia	Available	
106071650495	Book	Available	
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232557000000	DigitalMedia	Available	
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301162000000	DigitalMedia	Available	
323189472559	Book	Available	
337415000000	DigitalMedia	Available	
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405912000000	DigitalMedia	Available	
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444570000000	DigitalMedia	Available	
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491485519980	Book	Available	
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565393293913	Book	Available	
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636887000000	DigitalMedia	Available	
653070000000	DigitalMedia	Available	
658393445831	Book	Available	
705529635231	Book	Available	
730795000000	DigitalMedia	Available	
767151867182	Book	Available	
770388000000	DigitalMedia	Available	
771900436262	Book	Available	
778084897890	Book	Available	
785782374802	Book	Available	
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935982000000	DigitalMedia	Available	
946002405289	Book	Available	
963484000000	DigitalMedia	Available	
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14037923478...	Magazine	Available	
18020094731...	Magazine	Available	
18446744073...	Magazine	Available	

1 • SELECT * FROM borrowtransaction

2

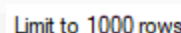
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1003	2024-01-20	2024-02-03	NULL	18446744073709551615	392888992260	384994027174
1004	2024-01-22	2024-02-05	2024-02-01	18020094731993304080	926509076716	192707852656
1005	2024-01-25	2024-02-08	NULL	730795000000	777315144757	482225930755
1006	2024-01-27	2024-02-10	NULL	232557000000	670019806533	847157462649
1007	2024-01-30	2024-02-13	2024-02-12	31515362208	106325369465	655919975680
1008	2024-02-02	2024-02-16	NULL	963484000000	862536141388	18984280546
1009	2024-02-05	2024-02-19	NULL	213939000000	908358000255	841606709552
1010	2024-02-07	2024-02-21	2024-02-20	770388000000	664904820126	63911658753
1011	2025-03-01	2025-03-15	NULL	491485519980	242955608655	277043089914
1012	2025-03-03	2025-03-17	NULL	323189472559	98844556456	384994027174
1013	2025-03-05	2025-03-19	NULL	771900436262	600450083709	277043089914
1014	2025-03-07	2025-03-21	NULL	658393445831	72458122216	384994027174
1015	2025-03-09	2025-03-23	NULL	139441916268	368798545706	277043089914
1016	2025-03-11	2025-03-25	2025-03-22	411437495765	345462137506	384994027174
1017	2025-03-13	2025-03-27	2025-03-24	984901710141	534771852898	277043089914
1018	2025-03-15	2025-03-29	2025-03-26	946002405289	588022882666	384994027174
1019	2025-03-17	2025-03-31	2025-03-28	106071650495	386979565044	277043089914
1020	2025-03-19	2025-04-02	2025-03-30	354023658054	609741931457	384994027174
1021	2025-03-21	2025-04-04	2025-04-01	565393293913	963447298338	277043089914
1022	2025-03-23	2025-04-06	2025-04-03	426357542125	152491468918	384994027174
1023	2025-03-25	2025-04-08	2025-04-05	488808576803	65543315395	277043089914
1024	2025-03-27	2025-04-10	2025-04-07	767151867182	151534339699	384994027174
1025	2025-03-29	2025-04-12	2025-04-09	785782374802	363714941482	277043089914
















1 SELECT * FROM client



Filter Rows:


Export/Import: 

Wrap Cell Co

	ClientID	Name	ContactInformation	Membership	AccountStatus	MembershipTypeID
▶	44756014165	Timothy Garza	(513) 758-0660	C	Warning	2
	65543315395	Elizabeth Vincent	NULL	C	Inactive	2
	72458122216	Angel Campbell	(759) 459-9246	C	Warning	2
	80444014843	Monique Santana	(001) 045-0932	C	Warning	2
	98844556456	Antonio Mcdonald	(116) 345-7923	C	Suspended	2
	100708262468	Justin Roberts	NULL	A	Inactive	0
	106325369465	Amanda Zavala	(711) 220-1868	A	Inactive	0
	108706256089	Brooke Mccarthy	(173) 864-0140	B	Suspended	1
	136178644349	Steven Henderson	(161) 162-0766	B	Active	1
	151534339699	Elizabeth Houston	(611) 877-5517	C	Active	2
	152491468918	Brian Cross	NULL	C	Active	2
	158774376402	Stephanie Williams	(089) 933-1886	A	Inactive	0
	162788563211	Danielle Browning	(001) 924-1157	C	Warning	2
	227259652028	Briana Smith	NULL	C	Suspended	2
	240123516435	Lorraine Garcia	(153) 671-0979	C	Inactive	2
	242955608655	Robert Giles	(869) 141-3145	A	Inactive	0
	320195254711	Megan Hernandez	(609) 767-0172	C	Suspended	2
	331180881681	Rodney Farrell	(873) 176-4303	C	Active	2
	345462137506	Melissa Moore	(769) 606-9602	C	Warning	2
	363714941482	Nicholas Kline	(961) 113-3060	B	Active	1
	368798545706	Sandra Miller	NULL	C	Active	2
	386979565044	William Garcia	(104) 714-2851	A	Warning	0
	392888992260	Christine Tran	(840) 801-6097	C	Suspended	2
	424533559245	Michelle Roman	(180) 132-0407	C	Warning	2
	487380045830	Miss Shelly Daniel...	(555) 082-4926	B	Active	1
	534771852898	David Moore	(789) 007-5470	A	Active	0
	540877402763	Tristan Davis	NULL	B	Warning	1
	574947128222	Mark Webb	(407) 970-3414	C	Inactive	2
	588022882666	Mrs. Meghan Ste...	(008) 913-1934	A	Inactive	0
	600450083709	Autumn Morse	(120) 769-8456	B	Suspended	1
	609741931457	Michael Grant	(909) 776-5823	A	Active	0
	616208160855	Ashley Ellis	(821) 972-9668	A	Suspended	0
	664904820126	Alyssa Reid	(191) 615-1090	C	Warning	2
	670019806533	Thomas Harris	(189) 471-9659	C	Warning	2
	690064463489	Christopher Hern...	(175) 868-8091	A	Active	0
	702703408825	Emily Sanchez	NULL	C	Active	2
	774490855370	Steven Butler	(699) 300-2489	A	Active	0
	777315144757	Eddie Martinez	NULL	A	Active	0
	779102014641	Norma Fisher	(876) 475-9382	C	Inactive	2
	799911506282	Rachel Klein	(615) 349-2635	A	Active	0
	862536141388	Stephanie Lawrence	(947) 751-5917	A	Active	0
	878829203004	Terri Turner	(660) 223-4500	B	Inactive	1
	881493143287	Daniel Rogers	(485) 291-1894	B	Warning	1
	908358000255	Autumn Robinson	(560) 123-0989	C	Suspended	2
	913800456057	Brian Rodriguez	(127) 626-6227	C	Warning	2

```
1 • SELECT * FROM book
2
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4
```

Result Grid								
Filter Rows:								
Edit: Export/Import: Wrap Cell Content:								
	ISBN	Title	Author	Publisher	Genre	PublicationYear	AvailabilityStatus	CopyID
▶	1023832279531	Twilight Covenant	Orion Black	Wandering Word Works	Mystery	2011	No	488808576803
	1273754815743	Shadows of Eternity	Elena Winters	Silver Quill Press	Fantasy	1974	No	491485519980
	3233694870907	Labyrinth of Souls	Mira Langley	Stormwatch Publications	Biography	1972	No	426357542125
	3262456709743	The Crystal Code	Marcus Thorn	Midnight Lantern Publishing	Fiction	1980	No	323189472559
	3621070726942	The Hollow Crown	Evelyn Storm	Crimson Hearth Books	Romance	1996	No	705529635231
	4029656002307	Echoes of Tomorrow	Julian Frost	Ironbound Books	Non-Fiction	1967	Yes	658393445831
	4887721181798	Whispers in the Wind	Isabella Drake	Starlight House	Biography	1968	No	771900436262
	5407350920242	Sands of Time	Ivy Nightshade	Golden Atlas Publishing	History	1981	No	785782374802
	5421207680208	Beneath the Iron Sky	Natalie Crowe	Echo Peak Publishing	Fiction	2012	Yes	778084897890
	5665528842083	Chronicles of the Void	Connor Vance	Emberlight Publications	History	1997	Yes	984901710141
	5976911093807	Frozen Reverie	Dorian Cross	Moonrise Editions	Science Fi...	2022	No	530239166629
	6030091410282	The Last Ember	Seraphina Moore	Northern Gate Books	Biography	1966	Yes	946002405289
	6684668056401	Legacy of the Stars	Tristan Vale	Zephyrbound Press	Fantasy	1963	Yes	550685652433
	672027792524	Voices from the Deep	Victor Stone	Twilight Beacon	Romance	1974	No	565393293913
	6796154824223	Path of the Ancients	Aria Tempest	Radiant Leaf Press	Romance	1987	No	354023658054
	6851259388489	The Ninth Horizon	Clara Voss	Sapphire Tome House	History	2024	No	767151867182
	7721818335132	Realm of Secrets	Lydia Blackwood	Whispering Pines Press	Fantasy	2024	No	411437495765
	8049194392752	The Forgotten Kingdom	Damien Hale	Celestial Ink	Mystery	2010	No	139441916268
	8769680377543	Oath of Ashes	Rowan Vale	Phoenix Crest Press	Fantasy	1986	No	925634441473
	9158736461602	Dreams of Glass	Gideon Stroud	Obsidian Scroll	Non-Fiction	1984	No	106071650495

Lim

```
1 • SELECT * FROM fine
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```

Result Grid				
Filter Rows:				
Edit:				
	FineID	TransactionID	Amount	PaymentDate
▶	101	1003	5.75	2023-05-20
	102	1005	3.25	2023-05-13
	103	1009	7.50	2023-11-18
	104	1001	0.00	2024-03-01
	105	1002	0.00	2024-05-27
	106	1004	0.00	2024-10-25
	107	1006	0.00	2025-01-18
	108	1007	0.00	2025-03-30
	109	1008	0.00	2025-04-16
	110	1010	0.00	2025-02-28
*	NULL	NULL	NULL	NULL

Functionality Testing

The screenshot shows the MySQL Workbench interface. The top menu bar includes File, Edit, View, Query, Database, Server, Tools, Scripting, and Help. The left sidebar contains the Navigator pane with a tree view of the database structure: CHEMAS, Filter objects, bookstore, bookstore_database, Tables, Views, Stored Procedures, Functions, librarydb, and sys. The main editor window displays a SQL query in the 'library_insert_data' tab. The query is as follows:

```

1 SELECT c.Name AS client_name,
2       b.CheckoutDate, b.DueDate,
3       f.Amount AS fine_amount, f.PaymentDate
4 FROM borrowtransaction b
5 JOIN client c ON b.ClientID = c.ClientID
6 JOIN fine f ON b.TransactionID = f.TransactionID
7 WHERE b.ReturnDate IS NULL
8       AND b.DueDate < CURDATE()
9       AND f.Amount > 0;

```

Below the query editor, the 'Result Grid' pane shows the results of the query. The grid has columns: client_name, CheckoutDate, DueDate, fine_amount, and PaymentDate. The results are as follows:

client_name	CheckoutDate	DueDate	fine_amount	PaymentDate
Christine Tran	2024-01-20	2024-02-03	5.75	
Eddie Martinez	2024-01-25	2024-02-08	3.25	2024-02-20
Autumn Robinson	2024-02-05	2024-02-19	7.50	

On the right side of the interface, there is a sidebar with the 'SQL Additions' pane and a 'Results' pane. The 'Results' pane contains a vertical toolbar with icons for 'Result Grid', 'Form Editor', 'Field Types', and 'Query Stats'.

MySQL Workbench

Local instance MySQL8 - VV...
File Edit View Query Database Server Tools Scripting Help

Navigator
Filter objects
bookstore
bookstore_database
Tables
Views
Stored Procedures
Functions
library
sys

Schemas

```

-- 查找所有未归还借书记录的客户
2 SELECT c.Name, c.ContactInformation, c.AccountStatus,
3         b.CheckoutDate, b.DueDate,
4         cp.Type, cp.AvailabilityStatus
5 FROM client c
6 JOIN borrowtransaction b ON c.ClientID = b.ClientID
7 JOIN copy cp ON b.CopyID = cp.CopyID
8 WHERE b.ReturnDate IS NULL;

```

Result Grid | Filter Rows | Export | Wrap Cell Contents

Name	ContactInformation	AccountStatus	CheckoutDate	DueDate	Type	AvailabilityStatus
Nancy Fisher	(919) 479-9382	Inactive	2024-01-15	2024-01-29	Book	Available
Christine Tran	(940) 801-6097	Suspended	2024-01-20	2024-02-03	Magazine	Available
Eddie Martinez		Active	2024-01-25	2024-02-08	DigitalMedia	Available
Thomas Harris	(386) 471-9649	Pending	2024-01-27	2024-02-10	DigitalMedia	Available
Stephane Lawrence	(947) 751-5917	Active	2024-02-02	2024-02-16	DigitalMedia	Available
Ashleen Robinson	(360) 223-0999	Suspended	2024-02-05	2024-02-19	DigitalMedia	Available

Administration Schemas Information

No object selected

Output

Action Output

Time Action Message Duration / Fetch

- 244 06:29:49 INSERT IGNORE INTO copy (CopyID, Type, AvailabilityStatus) VALUES ('014855','BOOK', 'Book', 'Available'), ('023149','ZSBB', 'Book', 'Available'), ('771900436262', 'Book', 'Available'), ('55203445831', 'Book', 'Availa... 47 rows(s) affected, 32 warnings: 1264 Out of range value for column 'CopyID' at row 21 1264 Out of range value for column 'CopyID' at row 23 1262 Duplicate entry '10446744072709551615' for key copy.PRIMARY... 0:00 sec
- 245 06:29:49 INSERT IGNORE INTO book (ISBN, Title, Author, Publisher, Genre, PublicationYear, AvailabilityStatus, CopyID) VALUES ('1237554815761', 'Shadows of Dementia', 'Gena Winters', 'Silver Owl Press', 'Fantasy', '1974', ... 20 rows(s) affected Records: 20 Duplicates: 0 Warnings: 0 0:01 sec
- 246 06:29:49 INSERT IGNORE INTO magazine (ID, Title, IssueNumber, PublicationDate, AvailabilityStatus, CopyID) VALUES ('91364862026376533587', 'Car and Driver', '93T', '2015-03-16', 'Available', '61203629632247482...', ... 20 rows(s) affected, 17 warnings: 1264 Out of range value for column 'CopyID' at row 3 1264 Out of range value for column 'CopyID' at row 4 1264 Out of range... 0:00 sec
- 247 06:29:49 INSERT IGNORE INTO digitalmedia (ID, Title, Creator, Format, ReleaseYear, Genre, AvailabilityStatus, CopyID) VALUES ('726774000000', 'Pianissimo', 'Robert Lurie', 'DVD', '2013', 'Thriller', 'In', '73075000000...', ... 20 rows(s) affected Records: 2 Duplicates: 0 Warnings: 0 0:00 sec
- 248 06:29:49 INSERT IGNORE INTO borrowtransaction (TransactionID, CheckoutDate, DueDate, ReturnDate, CopyID, ClientID) VALUES ('1001', STR_TO_DATE('01/15/2024', '%m/%d/%Y'), STR_TO_DATE('01/29/20...', ... 10 rows(s) affected, 1 warning: 1264 Out of range value for column 'CopyID' at row 3 Records: 10 Duplicates: 0 Warnings: 0 0:00 sec
- 249 06:29:49 INSERT IGNORE INTO fine (FineID, TransactionID, Amount, PaymentDate) VALUES ('1001', 1003, 5.75, NULL), ('102', 1005, 3.25, STR_TO_DATE('02/26/2024', '%m/%d/%Y')), ('103', 1009, 7.50, NULL), ('104', 1001...', ... 10 rows(s) affected Records: 10 Duplicates: 0 Warnings: 0 0:00 sec
- 250 06:29:49 INSERT IGNORE INTO reservation (ReservationID, ClientID, NameID, ReservationDate, Status) VALUES ('001', '779102019461', '779584897890', STR_TO_DATE('01/16/2024', '%m/%d/%Y'), 'Held'), ('002', '1627395...', ... 10 rows(s) affected, 1 warning: 1264 Out of range value for column 'NameID' at row 3 Records: 10 Duplicates: 0 Warnings: 1 0:00 sec
- 251 06:29:49 INSERT IGNORE INTO report (ReportID, DateOfGeneration, Type, StaffID) VALUES ('01120631512', '2016-05-12', 'Inventory', '277940009914', '09912180057', '2021-01-09', 'Inventory', '304994027174', '2006041121...', ... 5 rows(s) affected Records: 5 Duplicates: 0 Warnings: 0 0:00 sec
- 252 06:33:39 SELECT `copy`.`AS` AS `title\_name`, COUNT(*) AS `record\_count` FROM `copy` UNION ALL SELECT `book`.`COUNT(\*)` FROM `book` UNION ALL SELECT `magazine`.`COUNT(\*)` FROM `magazine` UNION ALL SELECT `dig... 7 rows(s) returned 0:00 sec / 0:00 sec
- 253 06:33:54 SELECT c.Name, c.ContactInformation, c.AccountStatus, b.CheckoutDate, b.DueDate, cp.Type, cp.AvailabilityStatus FROM client c JOIN borrowtransaction b ON c.ClientID = b.ClientID JOIN copy cp ON... 6 rows(s) returned 0:00 sec / 0:00 sec
- 254 06:34:03 SELECT c.Name AS client_name, b.CheckoutDate, b.DueDate, f.Amount AS fine_amount, f.PaymentDate FROM borrowtransaction b JOIN client c ON b.ClientID = c.ClientID JOIN fine f ON b.TransactionI... 3 rows(s) returned 0:00 sec / 0:00 sec
- 255 06:34:20 SELECT c.Name, c.ContactInformation, c.AccountStatus, b.CheckoutDate, b.DueDate, cp.Type, cp.AvailabilityStatus FROM client c JOIN borrowtransaction b ON c.ClientID = b.ClientID JOIN copy cp ON... 6 rows(s) returned 0:00 sec / 0:00 sec

Object Info Session

SQL Additions
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Read Only Context Help Snippets